Pinnacle Properties Group — Agent Experience Team: Technology Context

Services Owned by This Team

Service	Responsibility	Technology
Agent Service	Agent profile, lifecycle status, E&O and license record management, onboarding checklist	Node.js 20 + PostgreSQL 15
Recruiting Service	Pipeline management, prospect record, stage transitions, source attribution, Recruiting Admin queue	Node.js 20 + PostgreSQL 15
Onboarding Service	Checklist step tracking, step completion events, block-removal on 100% completion	Node.js 20 + PostgreSQL 15
Cap Tracking Service	Year-to-date cap accumulator, anniversary date management, projection engine	Node.js 20 + PostgreSQL 15
E&O Monitor Lambda	Nightly cron: check all E&O and license expiry dates; emit alerts and block events	Lambda (scheduled EventBridge rule)

Frontends Owned by This Team

Client	Technology	Notes
Agent Portal	React 18 SPA (TypeScript)	Cognito JWT auth; role-gated by agent status; responsive (desktop + tablet); Tailwind CSS
Branch Manager Dashboard	React 18 SPA (same codebase, role-gated views)	Multi-agent production view; cap progress report; E&O alert queue
Recruiting Admin Console	React 18 SPA (internal, same codebase)	Pipeline kanban, application review, license verification queue

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Key Technical Constraints

Stripe Connect for Agent Bank Accounts

- Each agent connects their bank account via **Stripe Connect Express** (OAuth flow from the agent portal)
- Stripe stores bank account details; Pinnacle never handles raw bank account numbers
- The Onboarding Service records the Stripe `accountId` against the agent profile on successful connection
- Stripe Connect payouts are initiated by the Disbursement Service (Transaction Management team), not this team — the Agent Experience team only manages the Connect onboarding flow

License Verification

- **TREC (Texas)**: TREC public license search API — Agent Service calls TREC API to verify license number + expiry date at onboarding; re-verified nightly by E&O Monitor Lambda
- **FREC (Florida)**: FREC does not provide a real-time API; verification is manual (Recruiting Admin uploads a screenshot of the FREC portal confirmation); the system stores the upload date as the verification date
- License data is stored in the Agent Service database; the nightly E&O Monitor Lambda compares stored expiry dates against today's date

E&O Monitor Lambda

- Runs at **6 AM CT daily** via EventBridge Scheduler
- Checks all agents with `status = ACTIVE` or `status = ONBOARDING` for E&O and license expiry dates
- Emits events: `agent.eo_warning` (at 30/14/7 days) and `agent.eo_expired` (on expiry date)
- `agent.eo_expired` event consumed by Agent Service (sets `transactionBlocked = true`) and Notification Service (sends email + SMS to agent and Branch Manager)

Cap Tracking

- Cap accumulator is updated by the Commission Engine (Transaction Management team) via an EventBridge event (`commission.brokerage_side.paid`) after each transaction closes
- Cap Tracking Service listens to this event and updates the running YTD accumulator
- Anniversary date is stored per agent; anniversary reset is triggered by the Cap Tracking Service on the anniversary date
- Projection engine: trailing 90-day average brokerage-side commission per month × remaining months to anniversary = projected additional contributions; estimated cap date calculated if trajectory continues

Calendly Integration (Showing Schedule)

- Phase 1: Calendly iFrame embed on the Listing Specialist's listing page in the agent portal
- Calendly event data is not synced to DealDesk (read-only embed only)
- Phase 2: Calendly API integration to sync showing times into the DealDesk transaction record

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External Integrations (this team)

Integration	Protocol	Direction	Notes
TREC API	REST	Agent Service → TREC	License verification at onboarding + nightly re-verification
Stripe Connect	OAuth 2.0 + REST	Agent Portal → Stripe	Bank account connection at onboarding; Stripe stores account details
DocuSign	REST + webhook	Onboarding Service ↔ DocuSign	Independent Contractor Agreement e-sign; webhook confirms completion
Follow Up Boss	REST (read-only)	Recruiting Service ← FUB	Legacy CRM sync; Phase 1 only; deprecated Phase 2
Calendly	iFrame (Phase 1) / REST API (Phase 2)	Agent Portal → Calendly	Showing schedule; Phase 1 embed only

MLS Sync Service	EventBridge (consumer)	MLS Sync → Agent Portal (listing status)	Listing status updates for Listing Specialists
SendGrid / Twilio	REST (via Notification Service)	Agent Service → Notification Service	E&O alerts, onboarding reminders, cap milestone notifications

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Non-Functional Requirements (this team)

Category	Requirement
E&O block accuracy	Block applied at 6 AM CT on expiry date; not retroactive; not delayed beyond 6 AM CT
Cap accumulator update	Updated within 5 minutes of `commission.brokerage_side.paid` EventBridge event
TREC verification	TREC API call responds within 5 seconds; timeout returns cached result with stale indicator
Onboarding checklist	Transaction creation blocked until all 6 steps have `status = complete` in Onboarding Service (enforced at Transaction Service, not just UI)
Stripe Connect flow	OAuth redirect + bank account confirmation completes within 3 minutes; on timeout, agent can retry without starting over
Dashboard performance	Agent dashboard loads within 2 seconds; branch manager multi-agent dashboard within 4 seconds (pagination for >50 agents)